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An active pole-to-ground fault location scheme for FB-MMC-MTDC system using the short-time blocking of sub-modules

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ABSTRACT

The pole-to-ground (PTG) fault current is small in the multi-terminal direct current (MTDC) system, which cannot be used to locate the fault. This paper proposes a PTG fault location scheme for MTDC based on full bridge modular multilevel converter (FB-MMC), by actively blocking the sub-modules (SMs). The SMs blocking generates a voltage pulse wave which travels between the MMC and the fault point. By using the wave polarity analysis method, the reflected wave from fault point can be identified. The improved complete ensemble empirical mode decomposition with adaptive noise (ICEEMDAN) is used to detect the reflected wave arrival time. The time difference between pulse start time and reflection arrival time is processed for fault location. A four-terminal MMC-MTDC simulation model established in PSCAD/EMTDC validates the proposed method have not affected the power transmission. Afterwards, the fault location accuracy under the influences of fault type, fault location and fault impedance issues are also confirmed.

1. Introduction

The direct current (DC) grid based on modular multilevel converter (MMC) offers a promising solution for the transmission and consumption of large-scale renewable energies [1]. Once a pole-to-ground (PTG) fault occurs in multi-terminal direct current (MTDC) system of symmetric monopole topology, all non-fault pole voltages increase double, threatening the safety of lines and equipment. After the fault isolation, fast and accurate fault location is needed in order to reduce the power outage time as well as the operating costs [2]. DC grounding mode or AC large resistance grounding mode is usually adopted in MMC-MTDC, and the equivalent impedance of PTG fault current loop is large. Thus, PTG fault current is small and the corresponding fault characteristics are not obvious, which cannot be used for fault location [3].

PTG fault location methods can be classified into three categories, the fault transient analysis methods [4–6], the travelling-wave methods [7], and the active injection methods [9–10]. The principle of fault transient analysis methods is using the wave equation and the electrical characteristics, and obtaining fault location from collected data [4]. In [5], researchers proposed a fault location scheme based on current limiting inductance voltage change rate. This scheme can reliably identify unipolar grounding fault, but cannot distinguish the exact fault pole. The principle of traveling-wave method is to calculate the wave propagation distance utilizing the arrival time of the wave front and the propagation velocity. Reference [7] proposed a fault location approach based on the analysis of head wave transient frequency and wave velocity, but the sampling frequency was too high. The above fault location schemes mostly depend on the information provided by the fault traveling wave, and the accuracy needs to be improved [8]. In the active injection scheme, the fault location can be calculated by analyzing the transient characteristics of the injected signal, or combining the travelling-wave method. Reference [9] presented a noniterative fault location technique using a probe power unit. This probe power unit can be implemented and executed by an intelligent electrical device for individual node. Research [10] presented a fault location method based on active harmonic injection, and an injection unit for active impedance estimation (AIE) is built and tested. However, the above active schemes need additional equipment, which lead to higher cost. When PTG faults occur, the converter control can be fully utilized for fault location. A single-terminal traveling wave location method by changing some sub-modules (SMs) status in a short time was proposed in reference [11]. However, this scheme applied only to the two-terminal system. Reference [12] suggested that the converter inject the trigonometric function signal with a specific frequency. A parameter identification model describing the correspondence between fault distance and transition resistance parameters is also constructed, which is used to locate the

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fault after the signal injection. Nonetheless, the frequency selection was greatly affected by the MTDC parameters in this scheme. Since the superposition of travelling waves along lines and converters is complex in MMC-MTDC, the accuracy of present active injection methods was greatly affected by MMC-MTDC topology and parameters. To address this issue, it is necessary to combine the high controllability of MMC and the analysis of MTDC characteristics to study a new active fault location scheme.

In the research of active injection method combining with travelling wave principle, accurate wave arrival time is one of the key points. Hence, it is necessary to use time–frequency analysis methods to extract effective arrival time features from the injection signal and avoid noise interference. The wavelet maximum theory is usually used to detect the wave arrival time [13]. However, it is difficult to choose the appropriate wavelet base to obtain the effective wavelet transform. Empirical mode decomposition (EMD) shows certain adaptability and sensitivity, but still suffers from the problem of mode aliasing [14]. Researches devoted more efforts to modify EMD by adding Gauss white noise, and proposed ensemble EMD (EEMD) [15] and complete EEMD (CEEMD) [16]. Nevertheless, the added Gauss white noise cannot be completely eliminated in EEMD, and the number of decomposed modes needs to be preset in CEEMD. Furthermore, Research [17] proposed an improved CEEMD with adaptive noise (ICEEMDAN) which showed the best accuracy in the family of EMD.

The main work of this paper is presenting an active fault detection and location scheme based on the high controllability of full bridge-MMC (FB-MMC). This paper firstly demonstrates the trigger method of the SMs, and analyzes the generated voltage pulse travelling process in MMC-MTDC. The reflected pulse is distinguished based on travelling analysis, and is detected by ICEEMDAN. In particular, this scheme can match different structure of MTDC without affecting the normal power transmission. No additional equipment is needed and the fault impedance proves to have no influence in this scheme. The rest of this paper is organized as follows: section II analyzes the topology of MMC-MTDC and the PTG fault characteristics. Section III analyzes the active voltage pulse generation and traveling process in MMC-MTDC. The pulse wave detecting method based on ICEEMDAN is presented in section IV. Section V is devoted to introducing the proposed fault location scheme, and section VI presents the simulation results under several scenarios. Finally, some conclusions are given in section VII.

2. The topology of MMC and MTDC system

Ring topology, radiate topology and net topology are the three most typical topologies of MTDC, which are all symmetric. Fig. 1 shows a four-terminal MMC-MTDC of ring topology. A full-bridge MMC includes three phases and six bridge arms, as shown in Fig. 2. The structure of each sub-module is also shown in Fig. 2. The SM has blocked status, inserted status and bypassed status [18].

The PTG fault transient characteristics are greatly affected by grounding mode. Under the DC-side grounding mode, the bridge arm SM capacitance and the fault point do not constitute a discharge circuit after

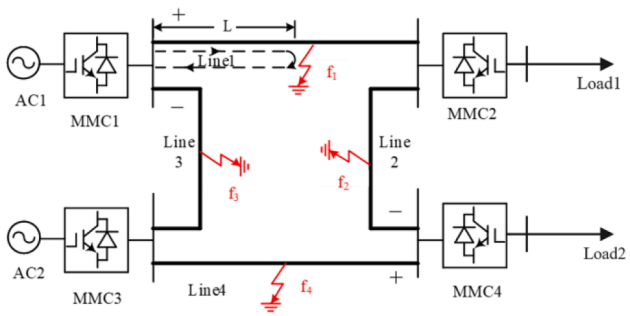


Fig. 1. A typical four-terminal MMC-DC network.

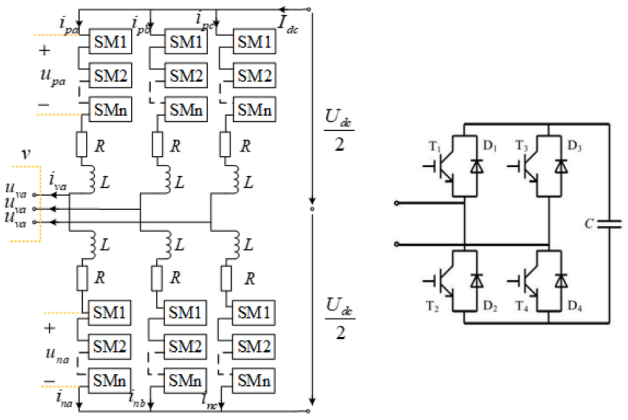


Fig. 2. Topological structure diagram of MMC.

the PTG fault. If a metal grounding fault occurs, the fault pole voltage drops to 0, and the non-fault pole voltage increases to twice the original voltage [19]. The converter output DC voltage is unchanged and the system power can also be transported normally.

AC side grounding mode contains transformer valve side through star reactor grounding mode and transformer neutral grounding mode. When a negative PTG fault occurs in the star reactor grounding mode, the corresponding bridge arm, the fault point and grounding point constitute the capacitor discharge circuit. The total resistance in the equivalent second-order circuit includes the reactor resistance and the fault impedance, which is generally non-negligible. The discharge transient process is an overdamped non-oscillation process and the fault current is small. In this process, the negative voltage reduces to 0, and the DC voltage remains unchanged under the control of the constant DC voltage controller. The positive voltage is doubled and the transmission power of the system is not affected [20]. When a PTG fault occurs in the neutral grounding mode, grounding loop and fault point form a discharge circuit. The SMs capacitors in the MMC fault pole arm are discharged, and the AC system is equivalent to a three-phase to ground fault through the fault point [21].

The grounding resistance has a direct impact on the DC PTG fault state in AC grounding mode. If the AC side is grounded with high impedance (greater than the critical damping), the equivalent circuit is approximately open circuit. The PTG short circuit has little effect on the system operation, and no protective action is required. If the AC side is grounded with low impedance, the discharge loop is underdamped, so the AC system and SM capacitance form oscillating discharge loop through the fault point. The bridge arm current is large, which may easily exceed the system protection setting value [22].

Therefore, MMC-MTDC generally adopts DC grounding mode or AC large resistance grounding mode considering the system safety. The PTG fault current amplitude is small due to the large equivalent resistance of the fault loop, which means the overcurrent protection of MMC will not be triggered. Meanwhile, the overvoltage protection action relay is about 1–3 s [23]. The system can be operated for a short time after PTG fault occurs. The MMC can be used for fault location in this time.

3. Voltage pulse generation and travelling wave propagation

3.1. The generation scheme of voltage pulse

As is known that the MTDC is usually required to run for a short period of time after PTG fault. Due to the negative level output capability, Full-bridge MMC system will not be blocked immediately after fault. MMC can make use of the switching of SMs for fault ride-through, ensuring that the active power and reactive power transmission are not interrupted. It also provides convenience for active injection fault location strategy.

DC voltage is supported by the SMs working in the inserted status. If a few of the SMs in the inserted or bypassed status of one phase are blocked shortly, the MMC output voltage will have a negative or positive fluctuation, which equals to a voltage pulse. This pulse is a wave propagating to the fault point and reflects. After measuring the time difference between the start time and the reflection arrival time, the distance between the converter and the fault point can be calculated. The specific short-time blocking scheme is shown in Fig. 3. In Fig. 3(a), the SMs are working in mode 2 and mode 5. The difference is the SMs input current direction. After the mode 2 SM is blocked, it will change to mode 1. The SM output voltage is still the capacitor voltage. After the mode 5 SM is blocked, it will change to mode 4, and the SM output voltage will change from the capacitor voltage to the diode conduction voltage, which close to 0. It is equivalent to inject a negative pulse into DC voltage. Similarly, in Fig. 3(b), the SMs in bypassed status also have two modes: mode 3 and mode 6. After the mode 3 SM is blocked, it will change to mode 1, and the SM output voltage will change from the conduction voltage of IGBT to the capacitor voltage. It is equivalent to inject a positive pulse into DC voltage. After the mode 6 SM is blocked, it will change to mode 4, and the SM output voltage remains unchanged. If the reference value of DC voltage is changed in the outer loop controller for a short time, the pulse will not occur on the DC line. This is because the existence of proportional integral link and inertial link in the outer loop controller will greatly weaken the slope of the DC voltage pulse.

A negative DC voltage pulse can be generated at the output of the converter by selecting several inserted status SMs with negative input current for short-time blocking. Similarly, a positive DC voltage pulse can be generated by selecting several bypassed status SMs with positive input current for short-time blocking. This DC voltage pulse is easy to detect even in the case of very short duration. Meanwhile due to the voltage support of the other two-phase bridge arm, the amplitude of this pulse accounts for a small proportion of the DC voltage. The system will quickly recover to be stable after fluctuation and the power transmission between converters will not be affected.

Such active detection method generated by power electronic equipment has certain requirements on the amplitude, frequency and duration of injected pulse [24]. The amplitude of voltage pulse should be higher than the fluctuation of DC voltage for detection. The fluctuation generally refers to the harmonic components generated by the DC voltage outer loop controller of the MMC station. Furthermore, it is recommended that the voltage pulse be within 20 % of the rated DC voltage [11]. In terms of frequency, the converter usually needs to generate voltage pulse for several times to obtain accurate fault distance calculation. The time interval of two adjacent voltage pulses should be longer than the switching time interval of SMs. The switching frequency of the converter SMs is generally 100 Hz, and the sampling frequency of the detection device is generally 10–50 kHz [25]. In addition, the trigger frequency should also be within the capacity range of the detection device [26]. In order to ensure that the two adjacent pulses are not aliased, the interval of the pulses should be over twice the propagation time of the traveling wave in the DC line. In terms of duration of each pulse, it should be determined according to the sampling frequency of the detection device. If the duration is too short, the detection device may not be able to capture the pulse signal. If the duration is too long, it means that some SMs are in a locked state for a long period of time, which may cause large DC voltage drop (or rise). Furthermore, the end and the head of the traveling wave may overlap, so the MMC controller period and the detection device sampling ability should be considered.

3.2. The wave propagation analysis

The DC voltage pulse wave propagates near the speed of light on the DC line. Fig. 4 is the schematic diagram of the traveling wave refraction and reflection. When the traveling wave propagates to the point where the wave impedance changes, the refraction and reflection will occur. Refraction wave, reflection wave and incident wave satisfy (1).

$$\begin{cases} u_{2q} = u_{1q} + u_{1f} \\ i_{2q} = i_{1q} + i_{1f} \\ i_{1f} = -u_{1f}/Z_1 \end{cases} \begin{cases} i_{1q} = u_{1q}/Z_1 \\ i_{2q} = u_{2q}/Z_2 \end{cases} \tag{1}$$

Where Z_1 and Z_2 are the wave impedance of two DC lines, u_{1q}, u_{1f}, u_{2q}, i_{1q}, i_{1f} and i_{2q} are the incident, reflected and refracted voltages and currents respectively. The relationship between these three voltages is shown in (2), where α is the refractive coefficient and β is the reflection coefficient.

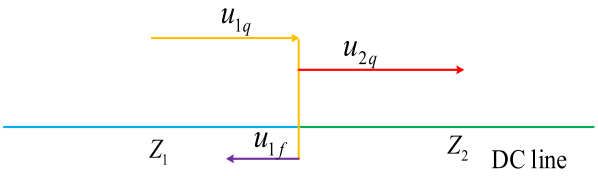


Fig. 4. Traveling refraction and reflection.

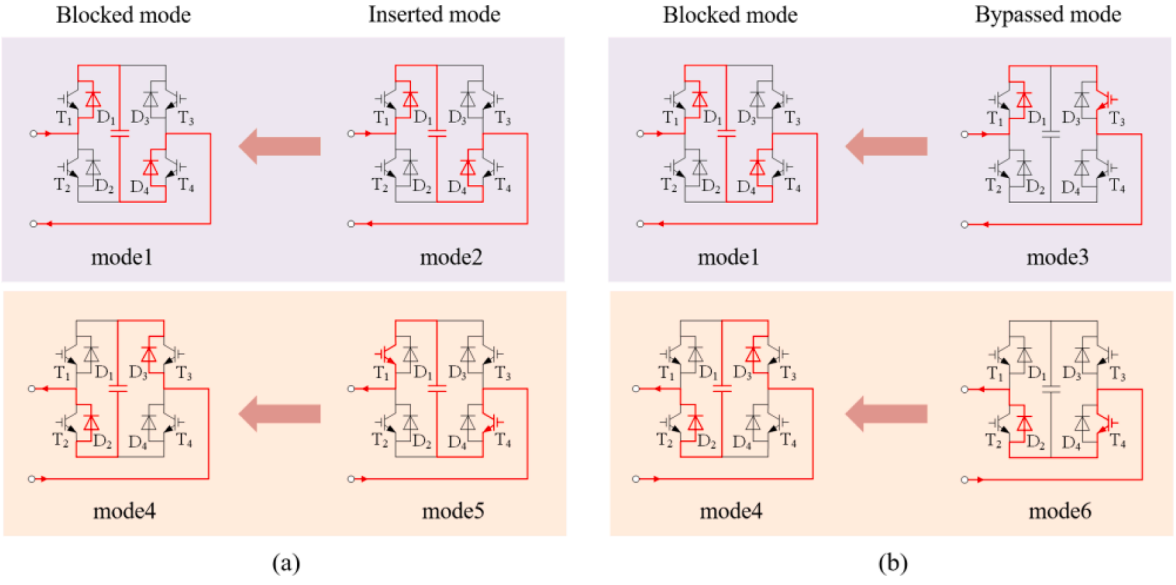


Fig. 3. SMs' state transition during the short time blocking (a)inserted to blocked status; (b)bypassed to blocked status.

$$\begin{cases} u_{2q} = \dfrac{2Z_2}{Z_1+Z_2}u_{1q} = \alpha u_{1q} \\ u_{1f} = \dfrac{Z_2-Z_1}{Z_1+Z_2}u_{1q} = \beta u_{1q} \end{cases} \tag{2}$$

$$\begin{cases} u_{2q} = \dfrac{Z_{c0}+4R_f}{Z_{c1}+Z_{c0}+4R_f} = \alpha u_{1q} \\ u_{1f} = \dfrac{-Z_{c1}}{Z_{c1}+Z_{c0}+4R_f} = \beta u_{1q} \end{cases} \tag{3}$$

As shown in Fig. 5, a PTG fault occurs at the distance of L from MMC1 to the DC line, while the fault impedance is R_f. However, the fault point cannot be decoupled with the transmission lines completely. The Karrenbauer transformation should be used when analyzing the travelling wave passing through the PTG fault point [27]. After phase-mode transforming, the refraction and reflection coefficients at the fault point are shown in (3). Where Z_{c1} and Z_{c0} are the one-mode and zero-mode wave impedance of the DC line, respectively. It can be seen that the refraction coefficient is $0 < \alpha < 1$ and the reflection coefficient is $\beta < 0$. Therefore, when the current traveling wave passes through the PTG fault point, the polarity of the refracted wave does not change, which is in negative direction. The amplitude is lower than that of the forward wave. While the polarity of the reflected wave changes. Fig. 5(a) shows the refraction and reflection of DC voltage pulse on the fault point and MMC1.

When the reflected wave reaches MMC1, A new refraction occurs, as shown in Fig. 5(b). The relationship between the reflected wave and the incident wave is shown in (4), where Z_{MMC} is the impedance of MMC1.

$$u_{3q} = \frac{2Z_{MMC}//Z}{Z_{MMC}//Z+Z}u_{1f} = \delta u_{1f} \tag{4}$$

According to the Peterson law and the equivalent circuit of MMC [28], there will be:

$$U_{3q}(s) = 2U_{1f}(s)\frac{s^2\cdot(\frac{2}{3}L_0)+\frac{N}{6C_0}}{s^3\cdot(\frac{2}{3}L_0)+s^2\cdot Z+s\cdot\frac{N}{6C_0}} \tag{5}$$

Where L_0 and C_0 represent the inductance of the bridge arm and the capacitance of the SMs. N represents the number of SMs on one bridge arm. Considering that the maximum coefficient of molecular denominator in (5) is much smaller than that of other terms, the (5) can be transformed into time domain without considering the influence of quadratic coefficient, as shown in (6). It can be seen that when $t > 0$, the polarity of u_{3q} and u_{1f} are the same. Therefore, the traveling wave from the fault point is detected on MMC1 in the positive direction.

$$u_{3q} = 2u_{1f}\cdot(1-e^{-\frac{t}{CZ}}) \tag{6}$$

By measuring the first and second occurrence time of DC voltage pulse at MMC exit, the fault distance L can be calculated by time difference and the wave velocity. When the single pole DC voltage is used for fault location, the coupling between the two DC lines will have a certain impact on the location results. A matrix decoupling the phase mode of the DC line is used in order to eliminate the coupling effect between the two poles [30]:

$$\begin{cases} S = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \\ [u_1\ u_0]^T = S^{-1}[u_p\ u_n]^T \end{cases} \tag{7}$$

Where u_1 and u_0 are one-mode voltage and zero-mode voltage, u_p and u_n are positive voltage and negative voltage respectively. Since one-mode is more stable and can obtain accurate wave polarity [29], this paper uses one-mode voltage for fault location. The wave velocity of one-mode can be measured with the whole distance of DC line under normal operation. The SM capacitor voltage does not change in the process of short-time blocking, and the MMC is no need to be charged after the wave velocity measurement.

4. Voltage pulse detection method based on ICEEMDAN

The key point of fault location scheme based on SMs short-time active blocking is how to find the arrival time of voltage pulse. The ICEEMDAN is used to measure the arrival time of pulse in this paper. The ICEEMDAN is modified based on CEEMDAN, and the difference is Gaussian white noise adding principle [31]. The decomposition process is as follows.

(1) Assuming a voltage series detected at the MMC export is u_{dc}. Add Gaussian white noise to the u_{dc}, and construct the mixed signal as:

$$u_{dc}^{(i)} = u_{dc} + \beta_0 E(w^{(i)})\ \ i \in \{1, \ldots, I\} \tag{8}$$

Where $w^{(i)}$ represents the added Gaussian white noise in the i-th turn, I is the number of the adding noise, and $E(w^{(i)})$ is the kth mode of the voltage series after EMD decomposition. Calculate the average of the local means to obtain the first residual:

$$R_1 = (N(u^{(i)}))\ \ i \in \{1, \ldots, I\} \tag{9}$$

Where $N(x^{(i)})$ is the Local mean value of new signal.

(2) Calculate the first mode as:

$$IMF_1 = u_{dc} - R_1 \tag{10}$$

(3) Add the Gaussian white noise again, and take stock of the second residual as the average of local means of the mixed signal $R_1 + \beta_1 E(w^{(i)})$. Calculate the second mode given by the following equation:

$$IMF_2 = R_1 - R_2 = R_1 - (N(R_1 + \beta_1 E(w^{(i)}))) \tag{11}$$

(4) Calculate the kth residual as:

$$R_k = (N(R_{k-1} + \beta_{k-1} E(w^{(i)}))) \tag{12}$$

The kth mode can be obtained:

$$IMF_k = R_{k-1} - R_k \tag{13}$$

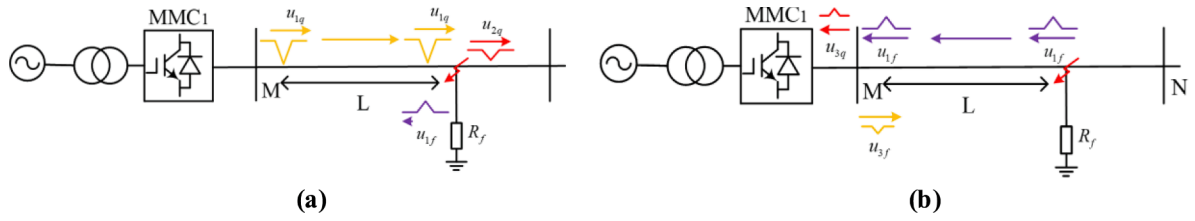


Fig. 5. DC voltage pulse refraction and reflection on fault point and MMC (a) refraction and reflection on fault point; (b) reflection on MMC.

(5) Repeat the step 4 and step 5 until the ICEEMDAN end condition is satisfied. The final residual can be seen as monotonic signal. As a result, the signal $x(n)$ can be represented by:

$$u_{dc}(n) = \sum_{m=1}^{M} IMF_m(n) + R_m(n) \tag{14}$$

Where M is the number of the modes calculated by ICEEMDAN. $IMF_m(n)$ is the intrinsic mode function (IMF) generated by ICEEMDAN decomposition. When the m goes from 1 to M, the IMF_m contains the signal components from high to low frequency. In order to accurately measure the arrival time of the DC voltage pulse, the DC line one mode voltage is first subjected to ICEEMDAN. The position where the maximum modulus appears in the IMF_m is the pulse arrival time. Since the duration of voltage pulse is short, we take the first IMF (IMF_1) for signal analysis in this paper, which represents the high frequency mode.

5. Pole-to-ground fault location scheme

5.1. Fault line determination strategy

It can be seen from Section II that after the PTG fault occurs in the symmetric MMC-MTDC, the fault pole voltage drops to close to 0, and the non-fault pole voltage doubles. When the PTP fault occurs, the positive and negative DC voltage amplitude remains equal due to the effect of grid grounding mode [32]. According to the DC voltage ratio of positive and negative lines, the occurrence of PTG fault can be judged.

$$\begin{cases} \dfrac{|U_{dcp}|}{|U_{dcn}|} > k_{setn}(a) \\ \dfrac{|U_{dcp}|}{|U_{dcn}|} < k_{setp}(b) \end{cases} \tag{15}$$

Where U_{dcp} is the positive DC line voltage, U_{dcn} is the negative DC line voltage. k_{set} is the set threshold. When the DC grid operates normally, the positive DC voltage is equal to the negative DC voltage. Considering a certain margin, this paper sets $k_{setn} = 1.2$, $k_{setp} = 0.8$. The fault characteristic satisfies (15) a is a negative PTG fault, and that satisfies (15) b is a positive PTG fault.

The traveling wave generated by the additional fault power supply travels from the fault point to both sides of the line [29]. When a positive PTG fault occurs at Line1, it is equivalent to adding an opposite voltage source according to the superposition theorem. At this time, there will be a step signal propagating to the four-terminal DC line. Since the distance from the fault point to each converter is different, the time for the step signal to reach each converter is different. When the step signal reaches a certain converter, the positive output voltage of the converter will decrease while the negative output voltage will increase. Normally, it can be seen from Fig. 1 and line parameters that the distance from the fault point to MMC1 and MMC2 is smaller than that to MMC3 and MMC4. Therefore, MMC1 and MMC2 detect the DC pulse earlier. After the PTG fault occurs, the fault line determination method is to detect the DC side voltage pulse time of each converter. The fault occurs in the line between two converters with the smallest voltage pulse detecting time. If the fault distance L is longer than the transmission line length L_1, it may be seen that the pulse is detected by MMC1 and MMC3 earlier. Hence, the fault pole should be determined firstly by (15), then only MMC1 and MMC4 need to be compared when a positive PTG fault occurs.

5.2. Fault location scheme

Accurate fault location is carried out after determining the fault type and line. If a PTG fault occurs at some point in Line1, a converter in either side of the fault point (such as the main station MMC1) actively generates a voltage pulse, propagating to the adjacent MMC2 and MMC3 in the speed of wave velocity v_1. The DC voltage pulse first reaches the fault point and reflects back to MMC1. The trigger time of MMC1 voltage pulse and the arrival time of the first reflected wave are detected. Active detection method can realize repeatable fault location so that can reduce the measurement error. The fault distance L is shown in (16).

$$L = \frac{1}{n}\sum_{i=1}^{n}\frac{(t_{2i} - t_{1i})}{2} \times v_1 \tag{16}$$

where t_{1i} and t_{2i} are the first reflected wave of the i-th DC voltage pulse trigger time and the first reflection arrival time; n is the number of voltage pulse; v_1 is the wave velocity of DC one-mode voltage, slightly slower than the light velocity [33]. The one-mode and zero-mode have different propagation wave speeds, so they need to be measured separately. According to the analysis in Section IV, the wave velocity of one-mode can be measured without affecting the normal operation of MTDC. In the case of normal operation of MMC-MTDC, the main station repeatedly launches active traveling wave. After measuring the arrival time of the first reflection wave front, the wave velocity of one-mode can be calculated:

$$v_1 = \frac{2L_{line}}{t_{reflection} - t_{start}} \tag{17}$$

where $t_{reflection}$ and t_{start} are the reflection arrival time and pulse start time in the velocity measurement procedure, respectively. L_{line} is the length of the whole line between two MMC.

6. Simulation verification

MMC-MTDC system model in Fig. 1 is built in PSCAD/EMTDC to verify the effectiveness of the proposed fault location scheme. The system is composed of main network with high voltage level and distribution network with low voltage level. The main network adopts master–slave control strategy. MMC1 acts as the main converter station to balance the active power, and the constant Udc and Q control are used to maintain the DC voltage to a constant value. MMC3 is connected to AC synchronous power grid, which can both be power sending and receiving ends, and is in a PQ control mode. MMC2 and MMC4 are connected to the wind farm, and the constant Uac and AC side frequency control strategy are used. A frequency-dependent phase domain model is used to construct DC line. The parameters of the four-terminal MMC-MTDC are shown in Table 1.

The DC grid enters the steady state at 0 s, and the DC line1 occurs a PTG fault at 0.1 s, and the main station MMC1 produces a DC voltage pulse at 0.2 s. Fig. 6 shows the output active power of the four-terminal converters at this time, and the normal exchange of the four-terminal active power can be obtained by simulation.

The step length of the simulation model in this paper is 0.2 μs, and the duration of the DC voltage pulse is set to 0.05 ms. The average switching frequency of the IGBT in MMC is defined as the opening

Table 1
System parameters of Four-terminal ring MMC-DC grid.

Items	Value	Items	Value
DC line voltage	±200 kV	SM capacitance	1105 μF
Rated AC voltageRated power	200 kV500 MW	Bridge arm inductance	386 mH
Converter level number	41	Capacitor voltage of SMs	10 kV
Line length	200 km	MMC1 control command	$U_{s1} = \pm 200$ kV. $Q_{MMC1} = 0$
Connection mode of transformer	YNd	MMC2 control command	$U_{s2} = 200$ kV.$f_{s2} = 50$ Hz
Equivalent leakage resistance of transformer	0.18 p.u.	MMC3 control commandMMC4 control command	$P_{MMC3} = 150$ MW. $Q_{MMC3} = 0U_{s4} = 200$ kV.$f_{s4} = 50$ Hz

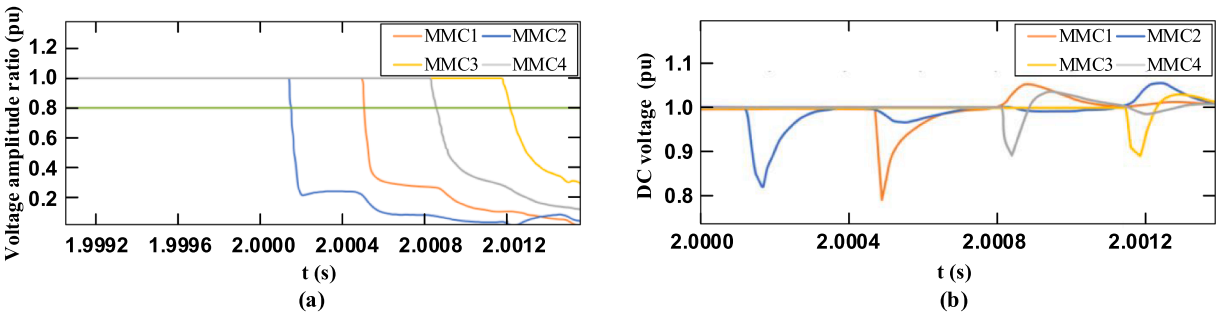


Fig. 6. MMC voltage after f1 (a) MMC output voltage amplitude ratio between poles; (b) MMC output voltage between poles.

numbers in 50 cycles:

$$f_{sw,ave} = \frac{\sum_{k=1}^{2N} n_{on,k}}{2N} \times 50 \tag{18}$$

Where $f_{sw,ave}$ is the average switching frequency of the IGBT, N is the number of SMs of the bridge arm, $n_{on,\ k}$ is the switching times of the kth IGBT in a power frequency cycle(50 Hz). The average switching frequency of the IGBT is within 100 Hz [25].

In this simulation, we select several SMs from one phase in bypassed status with positive current for short-time blocking to generate the positive DC voltage pulse. During the continuous action time of low-voltage pulse signal, the selected SMs may turn into inserted status. However, according to the above analysis, the DC voltage pulse duration is far less than the IGBT average switching period. Therefore, only a few SMs state will change in this duration time, the DC voltage pulse will not be affected.

6.1. Simulation of fault line determination

The positive PTG fault occurs at 150 km away from MMC1 on line1 at 2 s, and the fault impedance is 1 Ω. The positive and negative voltage amplitude of each converters is shown in Fig. 6(a). It can be seen that the positive and negative voltage amplitude ratio of each converter decreases rapidly, and is lower than the set value k_{setp} after the fault. The PTG fault of DC line can be judged by (15). The voltage between the positive and negative poles is shown in Fig. 6(b). The relationship between the distance from the fault point to each converter can be known: MMC2 < MMC1 < MMC4 < MMC3. MMC1 and MMC2 are the first converter to detect the voltage pulses, and the DC fault is located on the Line1.

Table 2 shows the fault location results under different fault conditions (different fault lines and fault distances). The simulation results show that the fault type and fault line can be reliably determined according to the proposed scheme.

6.2. Simulation of pole-to-ground fault location

The positive PTG fault occurs at 75 km away from MMC1 on Line1, and the fault impedance is set to 1 Ω. The fault occurred at2s, and MMC1 actively generates a negative DC voltage pulse at2.05 s. After the positive PTG fault occurs, four SMs from one phase in inserted status with negative current are selected for short-time blocking. During this period, the T_1 trigger signal and the output capacitor voltage of one SM are shown in Fig. 7. In order to show more intuitively, the output capacitor voltage of the SM is converted to the per unit value, and the reference value is the rated capacitor voltage of the SM. The T_1 trigger signal is 1 for the on signal, and 0 for the off signal. The blocking duration is 0.05 ms. During the short-time blocking period, the output capacitor voltage of the SM changes to 0. After the trigger pulse is unlocked, the output capacitor voltage of the SM is restored to the rated capacitor voltage.

The one-mode DC voltage of MMC1 is shown in Fig. 8. After the voltage pulse generates, the DC line voltage rises and quickly returns to normal value. The maximum rise value is 15 % of the rated DC voltage. When the traveling wave reaches the fault point, the reflection occurs and returns to MMC1 and the polarity of the traveling wave changes. The voltage pulse amplitude decreases a little, because the traveling wave has an attenuation effect during its propagation.

The trigger time of voltage pulse and the arrival time of the first reflection wave are determined by IMF_1 after ICEEMDAN. The detecting results are shown in Fig. 9. The two points of the maximum modulus in Fig. 10 are the starting time t_{1s} of the voltage pulse and the starting time t_{2s} of the reflected wave arriving at MMC1. The voltage pulse occurs every 0.1 s, in a total of 5 times, so the average time difference can be calculated as $\Delta t_{ave} = 5.03 \times 10^{-4}$ s.

In the normal operation state, the main station MMC1 generates the voltage pulse for the first time at2.05 s to measure the wave velocity. The pulse interval is 0.05 s, and the total number is 5 times. The one-mode DC voltage at MMC1 when the first pulse occurs is shown in Fig. 10. The maximum modulus of IMF1 is taken to detect the arrival time of the pulse wave front, as shown in Fig. 11. The corresponding moments of the two maximums in the figure are in chronological order: the start time of the transmitted pulse, the start time of the reflection pulse. We select the average wave velocity of each group as the pulse wave velocity, which is 2.9717×10^8 m/s. Substitution this pulse wave velocity into the equation (16), then, the fault distance can be calculated as 74.74 km. Furthermore, if a temporary fault occurs, the impedance characteristic at the fault point may not change after fault point recovery. Only a few pulses can be detected for fault location. To verify

Table 2
Fault detection results under different fault conditions.

Fault Line	Fault Type	Fault Location	$\lvert U_{dcp}/U_{dcn}\rvert$	MMC with the Smallest Time	Detection Result	Fault Line	Fault Type	Fault Location	$\lvert U_{dcp}/U_{dcn}\rvert$	MMC with the Smallest Time	Detection Result
Line1	Positive ground	25 km from MMC_1	$<k_{setp}$	MMC1,MMC2	L1 Positive ground	Line3	Positive ground	25 km from MMC_1	$<k_{setp}$	MMC1,MMC3	L3 Positive ground
Line1	Negative ground	75 km from MMC_1	$>k_{setn}$	MMC1,MMC2	L1 Negative ground	Line3	Negative ground	75 km from MMC_1	$>k_{setn}$	MMC1,MMC3	L3 Negative ground
Line2	Positive ground	25 km from MMC_2	$<k_{setp}$	MMC2,MMC4	L2 Positive ground	Line4	Positive ground	25 km from MMC_3	$<k_{setp}$	MMC3,MMC4	L4 Positive ground
Line2	Negative ground	75 km from MMC_2	$>k_{setn}$	MMC2,MMC4	L2 Negative ground	Line4	Negative ground	75 km from MMC_3	$>k_{setn}$	MMC3,MMC4	L4 Negative ground

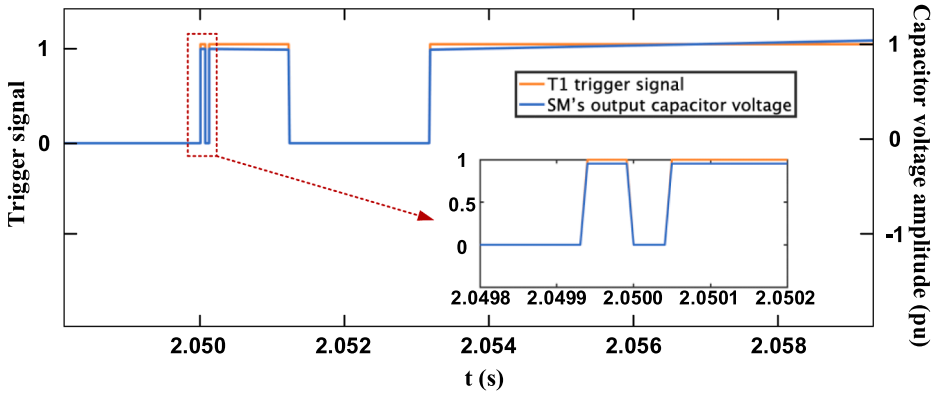


Fig. 7. T_1 trigger signal of a SM and the output capacitor voltage of the SM during the short-time blocking.

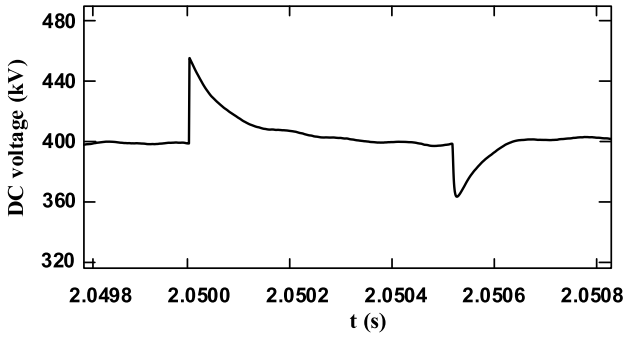


Fig. 8. The one-mode of MMC1 DC voltage.

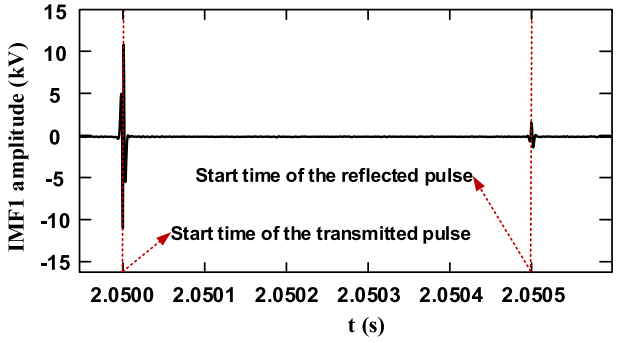


Fig. 9. DC voltage pulse trigger time and arrival time detected by IMF1 (fault state).

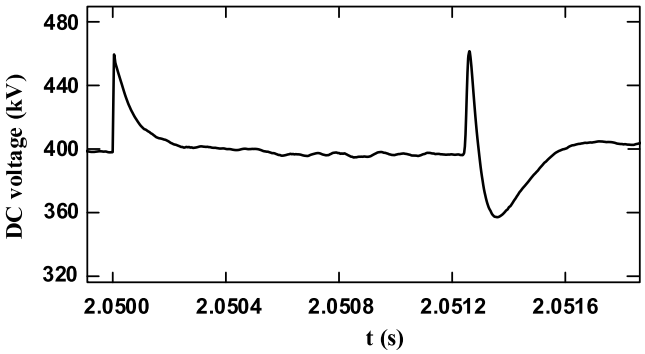


Fig. 10. MMC1 DC voltage's one-mode.

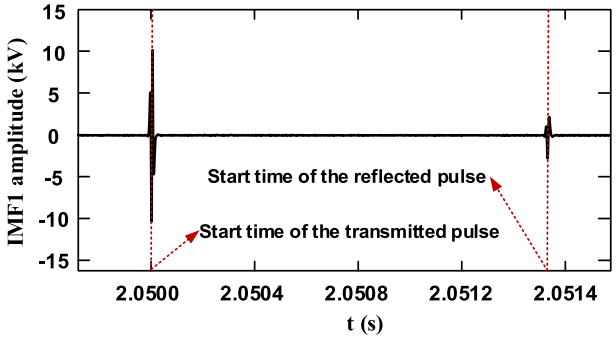


Fig. 11. DC voltage pulse trigger time and arrival time detected by IMF1 (normal state).

whether the method can be used for temporary fault, we choose only the first reflected pulse to calculate the fault distance.

The trigger time of voltage pulse and the arrival time of the first reflection wave are determined by IMF_1 after ICEEMDAN.

PTG faults are set in different DC lines and in different fault types. Table 3 shows the fault location results, and we calculate the fault location error according to (19). Finally, the correctness of the proposed scheme is verified by the simulation results.

$$\text{Error} = \frac{|l_{actual} - l_{calculation}|}{l_{actual}} \times 100\% \tag{19}$$

Where l_{actual} is the actual length between fault point and MMC, $l_{calculation}$ is the calculation result by proposed scheme.

6.3. Influence of fault impedance on fault location

For single conductor overhead line, the line impedance is 500 Ω, and for split conductor, its line impedance is generally 300 Ω. The fault impedance of DC PTG fault can generally reach hundreds of ohms [34]. According to (3) and (4), the fault impedance is the main factor affecting voltage pulse attenuation, which may influence fault location accuracy. When the fault impedance is small, $R_g//Z_c \ll Z_c$. From (3) the refraction coefficient $\alpha \approx 0$ and the reflection coefficient $\beta \approx -1$. When the DC voltage pulse propagates to the fault point, the reflection occurs approximately as total reflection. The converter can receive reflection information, which can be used for fault location.

When the fault impedance is large, the DC voltage pulse will propagate to the fault point and produce refraction. At the fault point, the refraction coefficient is $0 < \alpha < 1$ and the reflection coefficient is $\beta < 0$. There is still a reflection phenomenon so the converter can receive reflection information for fault location.

In summary, no matter in the case of large fault impedance or small

Table 3
Results of fault location.

Fault Line	Fault Type	Fault Location	Calculated Value	Error	Fault Line	Fault Type	Fault Location	Calculated Value	Error
L1	Positive permanent	25 km away from MMC1	25.07 km	0.28 %	L3	Positive permanent	25 km away from MMC1	25.11 km	0.44 %
L1	Negative permanent	75 km away from MMC1	74.74 km	0.35 %	L3	Negative permanent	75 km away from MMC1	75.46 km	0.61 %
L2	Positive permanent	25 km away from MMC2	24.98 km	0.08 %	L4	Positive permanent	25 km away from MMC3	25.03 km	0.12 %
L2	Negative permanent	75 km away from MMC2	74.35 km	0.87 %	L4	Negative permanent	75 km away from MMC3	74.61 km	0.52 %
L1	Positive temporary	25 km away from MMC1	25.09 km	0.36 %	L1	Positive temporary	75 km away from MMC1	74.41 km	0.79 %

fault impedance, the traveling wave will reflect at the fault point. The reflection coefficient β is <0, and the refraction coefficient of reflection wave at the converter station is greater than 0. Therefore, the DC voltage pulse detected at the converter is positive (assuming that the converter injects a negative voltage pulse). The different fault impedances values are set at the mid-point of Line1. And Table 4 shows the fault location results. It can be seen that the fault impedance has little effect on the proposed scheme.

7. Conclusion

This paper proposes an active pole-to-ground fault location scheme for FB-MMC-MTDC system using short-time blocking of SMs. Firstly, this scheme judges the fault type and fault pole by comparing the positive and negative DC voltages at each converter. The fault line can be quickly determined by detecting the arrival time sequence of the fault wave from each MMC. Since the converter will not be blocked in a short time, several inserted status or bypassed status SMs in one phase can be utilized for signal injection. By blocking the SMs for a short time, a DC voltage pulse is actively generated at the export of the MMC. The DC voltage pulse reflects at the fault point no matter how large the fault impedance is, and the propagation law of pulse in MMC-MTDC is analyzed. We use the ICEEMDAN to detect the traveling wave launch and arrival time, and the fault distance is accurately calculated by the time difference between them. This process can be repeated for accurate fault location. The simulation verification is carried out in a four-terminal FB-MMC-DC ring grid. The simulation results show that the proposed fault location scheme can accurately identify the fault location. The advantages of the proposed scheme are as follows:

(1) In the case of DC side fault in MTDC, this scheme can quickly and reliably distinguish the PTG fault and locate both permanent fault and temporary fault. The scheme can also match different structure of MTDC;
(2) Compared with the traditional fault location method that relies on the traveling wave produced by the fault point, this scheme can locate the fault repeatedly, and the location result is less affected by the fault impedance;
(3) When determining the fault line, only the sequence of time changes detected by the converters is needed, and there is no need for accurate time, so the communication requirements for multi-terminal converters are low.

CRediT authorship contribution statement

Guo Ziqing: Writing – original draft, Methodology, Software, Formal analysis. **Huang Wentao:** Conceptualization, Supervision, Resources, Funding acquisition. **Yu Moduo:** Investigation, Writing – review & editing. **Li Ran:** Validation, Data curation.

Table 4
Results of fault location with different fault impedance.

Fault Resistance	Fault Location	Calculated Value	Error
1 Ω	50 km away from MMC1	50.15 km	0.30 %
50 Ω	50 km away from MMC1	50.47 km	0.94 %
100 Ω	50 km away from MMC1	50.32 km	0.64 %

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

References

[1] Martinez-Rodrigo F, Ramirez D, Mendonça H, Pablo SD. MMC as nonlinear vector current source for grid connection of wave energy generation. Int J Electr Power Energy Syst 2019;113:686–98.
[2] Zhang L, Zou Y, Yu J, Qin J, Vittal V, Karady GG, et al. Modeling, control, and protection of modular multilevel converter-based multi-terminal HVDC systems: A review. CSEE J Power Energy Syst 2017;3(4):340–52.
[3] Dragicevic T, Lu X, Vasquez JC, Guerrero JM. DC Microgrids-Part II: A Review of Power Architectures, Applications, and Standardization Issues. IEEE Trans Power Electron 2016;31(5):3528–49.
[4] Xu J, Lü Y, Zhao C, Liang J. A model-based DC fault location scheme for multi-terminal MMC-HVDC systems using a simplified transmission line representation. IEEE Trans Power Delivery 2020;35(1):386–95.
[5] Li R, Xu L, Yao L. DC fault detection and location in meshed multiterminal HVDC systems based on DC reactor voltage change rate. IEEE Trans Power Delivery 2017; 32(3):1516–26.
[6] Yeap YM, Geddada N, Satpathi K, Ukil A. Time and frequency domain fault detection in a VSC-interfaced experimental DC test system. IEEE Trans Ind Inf 2018;14(10):4353–64.
[7] Duan J, Liu J, Lu H, Zhao Z. Fault location method based on traveling-wave instantaneous frequency for HVDC transmission lines. Proc CSEE 2016;36(7): 1842–8. https://doi.org/10.13334/j.0258-8013.pcsee.2016.07.010.
[8] Luis DADF. A brief history of fault location in transmission lines, in: *2019 6th IEEE History of Electrotechnology Conference* 2019: 44-47.
[9] Park J D, Candelaria J, MA L, Dunn K. DC ring-bus microgrid fault protection and identification of fault location. *IEEE Trans Power Del* 2013; 28(4): 2574-2584.
[10] Jia K, Bi T, Liu B, Christopher E, Thomas DWP, Sumner M. Marine Power Distribution System Fault Location Using a Portable Injection Unit. IEEE Trans Power Delivery 2015;30(2):818–26.
[11] Bi T, Wang S, Jia K. Single pole-to-ground fault location method for MMC-HVDC system using active pulse. IET Gener Transm Distrib 2018;12(2):272–8.
[12] Song G, Hou J, Guo B. Single-ended active injection for fault location in hybrid MMC-HVDC systems. International Journal of Electrical Power & Energy Systems 2021;124:106344.
[13] Liang R, Liu C, Peng N, Cheng M, Wang F. Fault location for power grid based on transient travelling wave data fusion via asynchronous voltage measurements. Int J Electr Power Energy Syst 2017;93:426–39.
[14] Wu Z, Huang NE. Ensemble empirical mode decomposition: A noise-assisted data analysis method. Adv Adapt Data Anal 2009;1(1):1–41.
[15] Wu Z, Huang NE. A study of the characteristics of white noise using the empirical decomposition method. Proc Royal Soc A 2004;460(2046):1597–611.

[16] Yeh JR, Shieh JS, Huang NE. Complementary ensemble empirical mode decomposition: a novel noise enhanced data analysis method. Adv Adapt Data Anal 2010;2(2):135–56.
[17] Marcelo AC, Gaston S, Maria ET. Improved complete ensemble EMD: A suitable tool for biomedical signal processing. Biomed Signal Process Control 2014;14:19–29.
[18] Xu J, Li J, Zhang J, Lu S, Jia X, Zhao C. Open-loop voltage balancing algorithm for two-port full-bridge MMC-HVDC system. Int J Electr Power Energy Syst 2019;109:259–68.
[19] Shamsoddini M, Vahidi B, Razani R, Mohamed YAI. A novel protection scheme for low voltage DC microgrid using inductance estimation. Int J Electr Power Energy Syst 2020;120:105992.
[20] Emhemed AAS, Burt GM. An Advanced Protection Scheme for Enabling an LVDC Last Mile Distribution Network. IEEE Trans Smart Grid 2017;5(5):2602–9.
[21] Zhao C, Li T, Yu L, Huang Y, Li L, Li X. DC pole-to-ground fault characteristic analysis and converter fault recovery strategy of MMC-HVDC. Proc CSEE 2014;34(21):3518–26. https://doi.org/10.13334/j.0258-8013.pcsee.2014.21.016.
[22] Acero DG, Cheah-Mane M, Páez JD, et al. Dc-MMC for the interconnection of HVDC grids with different line topologies. IEEE Trans Power Delivery 2021;37(3):1692–703.
[23] Tsovilis TE, Topcagic Z. DC overload behavior of low-voltage varistor-based surge protective devices. IEEE Trans Power Delivery 2020;35(5):2541–3.
[24] Hou J, Song G, Xu R, Peng C, Hussain KST. Fault property discrimination scheme in hybrid MTDC power system based on the amplitude distribution of injection signal. Int J Electr Power Energy Syst 2022;138:107930.
[25] Dong C, Ji P, Liu J, Shi P. MMC non-sorting equalizing strategy for reducing IGBT switching frequency. Power Electron 2020;54(9):119–22.
[26] Sun L, Huang W, Li R, Gao F, Tai N, Yu M. Capacity and Volume Balance of Buffering Converters for the Marine Pulsed Power System. IEEE Trans Ind Electron 2023;70(1):322–33.
[27] Lan T, Li Y, Duan X. High fault-resistance tolerable traveling wave protection for multi-terminal VSC-HVDC. IEEE Trans Power Delivery 2020;36(2):943–56.
[28] Ye H, Gao S, Li G, Liu Y. Efficient Estimation and Characteristic Analysis of Short-Circuit Currents for MMC-MTDC Grids. IEEE Trans Ind Electron 2021;68(1):58–269.
[29] Wang J, Zhang Y. Traveling wave propagation characteristic-based LCC-MMC hybrid HVDC transmission line fault location method. IEEE Trans Power Delivery 2021;37(1):208–18.
[30] Song G, Zhou D, Jiao Z, Suo N, Yun B. A new principle of fault location of DC transmission lines. Autom Electric Power Syst 2007;31(24):57–61.
[31] Huang D, Li S, Qin N, Zhang Y. Fault diagnosis of high-speed train bogie based on the improved-CEEMDAN and 1-D CNN algorithms. IEEE Trans Instrum Meas 2021;70(3508811):1–11.
[32] He J, Li B, Li Y, Qiu H, Wang C, Dai D. A Fast direction pilot protection scheme for multi terminal flexible DC power grid. Proc CSEE 2017;37(23):6878–87. https://doi.org/10.13334/j.0258-8013.pcsee.161573.
[33] Gao S, Suonan N, Song G, Zhang J, Jiao Z. Fault location method for HVDC transmission lines on the basis of the distributed parameter model. Proc CSEE 2010;30(13):75–80. https://doi.org/10.13334/j.0258-8013.pcsee.2010.13.011.
[34] Yang S, Xiang W, Yang R, He Y, Wen J. DC fault adaptive reclosing technology for flexible DC system based on half bridge MMC and hybrid DCCB. Proc CSEE 2020;40(14):4440–51. https://doi.org/10.13334/j.0258-8013.pcsee.190563.